

ENUMERATION OF DIVISIBLE DESIGN GRAPHS

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Deza graphs

Deza graphs were introduced in 1999 [1] as a generalisation of strongly regular graphs.

A k -regular graph on v vertices is called a **Deza graph** with parameters (v, k, b, a) , $b \geq a$ if the number of common neighbours of any two distinct vertices takes two values: a or b .

A Deza graph is called a **strictly Deza graph** if it has diameter 2 and is not strongly regular.

[1] M. Erickson, S. Fernando, W.H. Haemers, D. Hardy, J. Hemmeter, Deza graphs: A generalization of strongly regular graphs, *J. Comb. Des.*, **7** no. 6, (1999) 359–405

Enumeration of strictly Deza graphs

In 1999 the complete list of strictly Deza graphs with at most 13 vertices was presented (see [1]).

In 2011 an algorithm for enumeration of Deza graphs was presented by S. Goryainov and L. Shalaginov and this list was extended up to 16 vertices (see [2]).

In 2014 S. Goryainov and L. Shalaginov found all Cayley-Deza graphs with $a > 0$ up to 59 vertices and listed all corresponding groups (see [3, 4]).

In 2019 this algorithm was improved by S. Goryainov, D. Panasenko and L. Shalaginov and this list was extended up to 21 vertices (see [5]).

[2] S. V. Goryainov, L. V. Shalaginov, On Deza graphs with 14, 15, and 16 vertices, *Sib. Electron. Mat. Izv.*, **8**, (2011) 105–115 (in Russian)

[3] S. V. Goryainov, L. V. Shalaginov, Cayley-Deza graphs, on less than 60 vertices, *Sib. Electron. Mat. Izv.*, **11** (2014) 268–310 (in Russian)

[4] List of all Cayley-Deza graphs with $a > 0$ up to 59 vertices, http://alg.imm.uran.ru/dezagraphs/deza_cayleytab.html

[5] List of all strictly Deza graphs up to 21 vertices, <http://alg.imm.uran.ru/dezagraphs/dezatab.html>

Divisible designs

An incidence structure with constant block size k is a (group) **divisible design** whenever the point set can be partitioned into m classes of size n , such that two points from one class occur together in exactly λ_1 blocks, and two points from different classes occur together in exactly λ_2 blocks.

A divisible design D is called **symmetric** (SDD for short) if the dual of D (that is, the design with the transposed incidence matrix) is again a divisible design with the same parameters as D .

Divisible design graphs

A k -regular graph on v vertices is called a **divisible design graph** (DDG for short) with parameters $(v, k, \lambda_1, \lambda_2, m, n)$ if its vertex set can be partitioned into m classes of size n , such that two distinct vertices from the same class have exactly λ_1 common neighbors, and two vertices from different classes have exactly λ_2 common neighbors.

A divisible design graph is a graph whose adjacency matrix is the incidence matrix of a symmetric divisible design.

Note that a DDG with $m = 1$, $n = 1$, or $\lambda_1 = \lambda_2$ is a (v, k, λ) -graph. If this is the case, we call the DDG **improper**, otherwise it is called **proper**.

The definition implies that divisible design graphs are Deza graphs.

Feasible parameters of divisible design graphs

Divisible design graphs were first studied in master's thesis by M.A. Meulenberg [6] and the list of feasible parameters of divisible design graphs up to 50 vertices was presented.

In three following papers [7, 8, 9] feasible parameters of divisible design graphs up to 27 vertices were studied in more details and the existence of graphs was resolved in all but one cases.

[6] M.A. Meulenberg, Divisible Design Graphs, Master's thesis, *Tilburg University*, 2008.

[7] W.H. Haemers, H. Kharaghani, M.A. Meulenberg, Divisible Design Graphs, *J. of Comb. Th., Series A*, **118** (2011) 978–992

[8] D. Crnkovic, W.H. Haemer, More about Divisible Design Graphs, *CentER Discussion Paper*, **2011-140** (2011)

[9] D. Crnkovic, W.H. Haemer, Walk-regular divisible design graphs, *Designs, Codes and Cryptography*, **72** (2014) 165–175

Eigenvalues of divisible design graphs

Divisible Design Graphs graphs have at most 5 distinct eigenvalues.

Formulas for eigenvalues and their multiplicities are presented in the table:

Eigenvalues	Multiplicity	
$\theta_0 = k$	1	
$\theta_1 = \sqrt{k - \lambda_1}$	f_1	
$\theta_2 = -\sqrt{k - \lambda_1}$	f_2	$f_1 + f_2 = m(n - 1)$
$\theta_3 = \sqrt{k^2 - \lambda_2 v}$	g_1	
$\theta_4 = -\sqrt{k^2 - \lambda_2 v}$	g_2	$g_1 + g_2 = m - 1$

Searching for feasible parameters

For fixed v :

- (1) All possible number of classes m and class sizes n are calculated.
- (2) k runs from 3 to $v-3$, both λ_1 and λ_2 run from $\max(0; 2k-v)$ to k , with exclusion of the cases for which $\lambda_1 = \lambda_2$, $\lambda_1 = k$, $\lambda_1 = k-1$, $\lambda_2 = 0$, $\lambda_2 = 2k-v$ (for these cases, all possible divisible design graphs are known, see [7]).
- (3) For the remaining possibilities it is checked whether $k^2 - \lambda_2 v$ or $k - \lambda_1$ are squares. If one of them is not a square, then the corresponding multiplicities are equal. If both of them are not squares, then the corresponding parameters are excluded.

Searching for feasible parameters

(4) The following cases are excluded [4]:

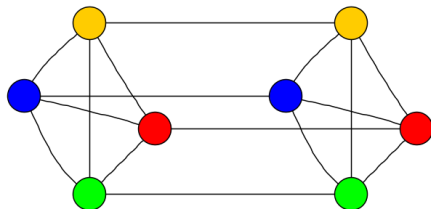
- $\lambda_1 = k - 1$ and n is odd
- $k^2 - \lambda_2 v = k - \lambda_1$
- $k^2 = k + \lambda_1(n - 1) + \lambda_2 n(m - 1)$
- $f_1 = f_2$ and $g_1 = g_2$
- $f_1 = f_2$ or $g_1 = g_2$ are not integers
- If only one of $k^2 - \lambda_2 v$ or $k - \lambda_1$ is not a square, the yet undetermined multiplicities can be computed. The case when they are not integers is excluded.

Equitable t -partition of a graph

Let $V_1 \cup V_2 \cup \dots \cup V_t$ be the partition of the vertex set of a graph Γ with the property that every vertex of V_i has exactly r_{ij} neighbours in V_j .

Then $V_1 \cup V_2 \cup \dots \cup V_t$ will be an equitable t -partition of Γ .

Matrix $R = (r_{ij})_{t \times t}$ is called the quotient matrix of equitable partition.



$(8, 4, 0, 2, 4, 2)$

$$R = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Enumeration of quotient matrices

In case of divisible design graphs, the partition into m classes induce equitable m -partition.

For the quotient matrix R of this partition the following properties hold:

- $0 \leq r_{ii} \leq n - 1$
- $0 \leq r_{ij} \leq n$, if $i \neq j$
- $\sum_{j=1}^m r_{ij} = k$
- $R^2 = RR^T = (k^2 - \lambda_2 v)I_m + \lambda_2 n J_m$, where I_m is an identity matrix of size m and J_m is an all-ones matrix of size m ([5]).

Enumeration of quotient matrices

For given parameters $(v, k, \lambda_1, \lambda_2, m, n)$:

- (1) We generate all possible r_{11} , $0 \leq r_{11} \leq n - 1$.
- (2) For all obtained r_{11} we generate all unordered partitions of $k - r_{11}$ as sum of $m - 1$ numbers. We use unordered partitions since the permutation of r_{1j} , $j \geq 2$ does not change the final result. For example, this two rows will give the same quotient matrix up to classes renumbering:

$$(1 \ 5 \ 5 \ 7), (1 \ 7 \ 5 \ 5).$$

- (3) We check if generated row satisfies the equality for R^2 . If not, we reject this row.
- (4) We repeat steps (1)-(3) for other rows, but we generate ordered partitions this time.
- (5) After we generate all possible i , $i \geq 2$ rows of the quotient matrix, we leave only non-equal up to classes renumbering options.

Enumeration of divisible design graphs

For given quotient matrix we generate all possible rows of the adjacency matrix, relying on known partition into classes.

We say that two partially filled matrices are equivalent if the graphs determined by them are isomorphic with respect to partition into classes.

After addition of a new row of the adjacency matrix, all partial matrices were checked for equivalence and only nonequivalent were considered for adding the next row.

Enumeration results

v	k	λ_1	λ_2	m	n	θ_1^{f1}	θ_2^{f2}	θ_3^{g1}	θ_4^{g2}	Total
8	4	0	2	4	2	2^1	-2^3	0^3	$-$	1!
12	5	0	2	6	2	$\sqrt{5}^3$	$-\sqrt{5}^3$	$-$	-1^5	1!
12	5	1	2	4	3	2^2	-2^6	1^3	$-$	1!
12	6	2	3	3	4	2^3	-2^6	0^2	$-$	1!
12	7	3	4	4	3	2^2	-2^6	1^2	-1^1	1!
15	4	0	1	5	3	2^5	-2^5	$-$	-1^4	1!
18	9	6	4	6	3	$\sqrt{3}^6$	$-\sqrt{3}^6$	3^1	-3^4	1!
20	7	3	2	4	5	2^4	-2^{12}	3^3	$-$	1!
20	9	0	4	10	2	3^4	-3^6	1^3	-1^6	1!
20	9	0	4	10	2	3^5	-3^5	$-$	-1^9	1!
20	13	9	8	4	5	2^4	-2^{12}	3^2	-3^1	1!
24	6	2	1	3	8	2^9	-2^{12}	$\sqrt{12}^1$	$-\sqrt{12}^1$	1!
24	7	0	2	8	3	$\sqrt{7}^8$	$-\sqrt{7}^8$	$-$	-1^7	1!
24	8	4	2	4	6	2^5	-2^{15}	4^3	$-$	1!
24	8	4	2	4	6	2^7	-2^{13}	4^2	-4^1	1!
24	8	4	2	4	6	2^9	-2^{11}	4^1	-4^2	6!
24	10	2	4	12	2	$\sqrt{8}^6$	$-\sqrt{8}^6$	2^3	-2^8	5!
24	10	3	4	8	3	$\sqrt{7}^8$	$-\sqrt{7}^8$	2^1	-2^6	2!
24	10	6	3	3	8	2^8	-2^{13}	$\sqrt{28}^1$	$-\sqrt{28}^1$	1!
24	14	6	8	12	2	$\sqrt{8}^6$	$-\sqrt{8}^6$	2^2	-2^9	1!
24	14	7	8	8	3	$\sqrt{7}^8$	$-\sqrt{7}^8$	$-$	-2^7	1!
24	16	12	10	4	6	2^5	-2^{15}	4^2	-4^1	1!
24	16	12	10	4	6	2^7	-2^{13}	4^1	-4^2	1!
24	16	12	10	4	6	2^9	-2^{11}	$-$	-4^3	4!

Enumeration results

v	k	λ_1	λ_2	m	n	θ_1^{f1}	θ_2^{f2}	θ_3^{g1}	θ_4^{g2}	Total
27	8	4	2	9	3	2^7	-2^{11}	$\sqrt{10}^4$	$-\sqrt{10}^4$	1!
27	18	9	12	9	3	3^6	-3^{12}	0^8	—	2!
28	6	2	1	7	4	2^9	-2^{12}	$\sqrt{8}^3$	$-\sqrt{8}^3$	1!
28	9	5	2	4	7	2^6	-2^{18}	5^3	—	1!
28	13	0	6	14	2	$\sqrt{13}^7$	$-\sqrt{13}^7$	—	-1^{13}	1!
28	13	4	6	7	4	3^9	-3^{12}	1^1	-1^5	16!
28	15	6	8	7	4	3^7	-3^{14}	1^6	—	56!
28	15	6	8	7	4	3^8	-3^{13}	1^3	-1^3	4!
28	19	15	12	4	7	2^6	-2^{18}	5^2	-5^1	1!
32	10	2	3	8	4	$\sqrt{8}^{12}$	$-\sqrt{8}^{12}$	2^1	-2^6	2!
32	10	6	2	4	8	2^7	-2^{21}	6^3	—	1!
32	10	6	2	4	8	2^{10}	-2^{18}	6^2	-6^1	1!
32	10	6	2	4	8	2^{13}	-2^{15}	6^1	-6^2	15!
32	14	2	6	16	2	$\sqrt{12}^8$	$-\sqrt{12}^8$	2^4	-2^{11}	15!
32	15	6	7	4	8	3^{12}	-3^{16}	—	-1^3	2+
32	16	0	8	16	2	4^6	-4^{10}	0^{15}	—	1!
32	17	8	9	4	8	3^{17}	-3^{11}	1^2	-1^1	?
32	18	6	10	16	2	$\sqrt{12}^8$	$-\sqrt{12}^8$	2^3	-2^{12}	1!
32	22	18	14	4	8	2^7	-2^{21}	6^2	-6^1	1!
32	22	18	14	4	8	2^{10}	-2^{18}	6^1	-6^2	1!
32	22	18	14	4	8	2^{13}	-2^{15}	—	-6^3	9!

Enumeration results

v	k	λ_1	λ_2	m	n	$\theta_1^{f_1}$	$\theta_2^{f_2}$	$\theta_3^{g_1}$	$\theta_4^{g_2}$	Total
35	12	3	4	7	5	3^{12}	-3^{16}	2^3	-2^3	$2!$
35	12	3	4	7	5	3^{14}	-3^{14}	—	-2^6	$3854!$
36	9	3	2	12	3	$\sqrt{6}^{12}$	$-\sqrt{6}^{12}$	3^4	-3^7	$3!$
36	9	4	2	18	2	$\sqrt{5}^9$	$-\sqrt{5}^9$	3^7	-3^{10}	$7!$
36	11	7	2	4	9	2^8	-2^{24}	7^3	—	$1!$
36	17	0	8	18	2	$\sqrt{17}^9$	$-\sqrt{17}^9$	—	-1^{17}	$1!$
36	24	15	16	4	9	3^{12}	-3^{20}	0^3	—	$3+$
36	25	21	16	4	9	2^8	-2^{24}	7^2	-7^1	$1!$
36	27	21	20	12	3	$\sqrt{6}^{12}$	$-\sqrt{6}^{12}$	3^1	-3^{10}	$1!$
38	9	0	2	19	2	3^8	-3^{11}	$\sqrt{5}^9$	$-\sqrt{5}^9$	$2!$

Construction 1

Suppose Γ is an antipodal distance-regular graph of diameter 3 with antipodal classes of size r . Denote by A_i the matrix of a relation ‘to be at distance i ’ on the vertices of Γ . If $\lambda = \mu + 2$, then matrix $A = A_1 + A_3$ is a matrix of a DDG with parameters $(r(2\mu + 4), 2\mu + r + 2, r - 2, \mu + 2, 2\mu + 4, r)$.

Example

Halved 6-cube is an antipodal distance-regular graph with antipodal classes of size 2 and $\lambda = \mu + 2$. Connecting antipodal vertices gives DDG with parameters $(32, 16, 0, 8, 16, 2)$.

New constructions

Let G be a k -regular graph on v vertices with the smallest eigenvalue $\lambda_{\min}$. A **Hoffman-coclique** is a coclique meeting the Hoffman upper bound $c = v\lambda_{\min}/(\lambda_{\min} - k)$.

A **Hoffman coloring** in G is a partition of the vertices into Hoffman-cocliques.

Construction 2

Suppose Γ is a strongly regular graph with parameters $(v, k, \mu + 2, \mu)$ and Γ has Hoffman coloring with Hoffman-cocliques of size n ($v = mn$). Let A be an adjacency matrix of Γ , in which Hoffman-cocliques are located on the main diagonal, $K = K_{(m,n)}$, $I = I_v$. Then matrix $A + K - I$ is the adjacency matrix of a DDG with parameters $(mn, k + n - 1, n + \mu - 2, \mu + \frac{2k}{m-1}, m, n)$.

Example

Consider triangular graph $T(8)$ and three Chang graphs with parameters $(28, 12, 6, 4)$. With construction 2 we can obtain 56 DDGs with parameters $(28, 15, 6, 8, 7, 4)$.

Construction 3

Let Γ be a DDG with parameters $(v, k, \lambda_1, \lambda_2, m, n)$ and quotient matrix $R = aI_m + b(J_m - I_m)$. Take s copies of Γ and label all blocks of canonical partition in each copy with numbers $1, \dots, m$. Then connect all vertices from blocks with the same label (adjacency inside the block does not change). The resulting graph is a DDG with parameters $(vs, k + (s-1)n, \lambda_1 + (s-1)n, \lambda_2, ms, n)$, if $\lambda_2 = 2b = 2a + (s-2)n$.

Construction 4

Let D be a symmetric divisible design with parameters $(v, k, \lambda_1, \lambda_2, m, n)$, such that every block contains the same number of points $\frac{k}{m}$ from each class. Γ is the incidence graph of D . Construct new graph Γ^* with the same vertex set as Γ and vertices x, y are adjacent in Γ^* when x, y are adjacent in Γ , or x, y are points from different classes of D , or x, y are blocks from different classes of dual design of D . Γ^* is a DDG with parameters $(2v, k + (m - 1)n, \lambda_1 + (m - 1)n, \lambda_2 + (m - 2)n, 2m, n)$, if $2(m - 1)\frac{k}{m} = \lambda_2 + (m - 2)n$.

Construction 5

Let D be a symmetric divisible design with parameters $(v, k, \lambda_1, \lambda_2, m, n)$ with even m , such that every block contains the same number of points $\frac{k}{m}$ from each class. Γ is the incidence graph of D . Split the set of classes of blocks and the set of classes of points into pairs. Construct a new graph Γ^* with the same vertex set as Γ and vertices x, y are adjacent in Γ^* when x, y are adjacent in Γ or x, y are from different classes of the same pair of classes partition. Γ^* is a DDG with parameters $(2v, k + n, \lambda_1 + n, \lambda_2, 2m, n)$, if $\frac{2k}{m} = \lambda_2$.

A **weighing matrix** $W(n, k)$ of order n and weight k is an $n \times n$ $(0, 1, -1)$ -matrix, such that $WW^T = kI_n$.

Construction 6

Let W be a $(4t, 4(t-1))$ -weighing matrix, such that the main diagonal of W contains blocks of zeros of size 4. Construct matrix A' by replacing each 0 with O_2 , each 1 with I_2 and each -1 with $J_2 - I_2$. Then matrix $A = A' + I_t \otimes ((J_4 - I_4) \otimes J_2)$ is the adjacency matrix of a DDG Γ with parameters $(8t, 4t + 2, 6, 2t + 2, 4t, 2)$.

Construction 7

Let Γ be a DDG obtained from Construction 20 with adjacency matrix A . The main diagonal of the A consists of $I_t \otimes ((J_4 - I_4) \otimes J_2)$, which gives a partition of Γ into complete multipartite graphs with 4 parts of size 2. Construct new graph Γ' by removing the edges of the complete bipartite subgraph $K_{4,4}$ from each part of this partition. Then Γ' is a DDG with parameters $(8t, 4t - 2, 2, 2t - 2, 4t, 2)$.

Unclassified DDGs

v	k	λ_1	λ_2	m	n	$\theta_1^{f_1}$	$\theta_2^{f_2}$	$\theta_3^{g_1}$	$\theta_4^{g_2}$	Total	Comment
27	8	4	2	9	3	2^7	-2^{11}	$\sqrt{10}^4$	$-\sqrt{10}^4$	1!	sporadic
28	6	2	1	7	4	2^9	-2^{12}	$\sqrt{8}^3$	$-\sqrt{8}^3$	1!	sporadic
28	13	4	6	7	4	3^9	-3^{12}	1^1	-1^5	16!	–
32	10	2	3	8	4	$\sqrt{8}^{12}$	$-\sqrt{8}^{12}$	2^1	-2^6	2!	structure
32	15	6	7	4	8	3^{12}	-3^{16}	–	-1^3	2+	Cayley
32	17	8	9	4	8	3^{17}	-3^{11}	1^2	-1^1	?	–
35	12	3	4	7	5	3^{12}	-3^{16}	2^3	-2^3	2!	–
36	9	3	2	12	3	$\sqrt{6}^{12}$	$-\sqrt{6}^{12}$	3^4	-3^7	3!	1 unclassified
36	24	15	16	4	9	3^{12}	-3^{20}	0^3	–	3+	Cayley
38	9	0	2	19	2	3^8	-3^{11}	$\sqrt{5}^9$	$-\sqrt{5}^9$	2!	–

The list of proper divisible design graphs up to 39 vertices is available by <http://alg.imm.uran.ru/dezagraphs/ddgtab.html>. This web page provides access to adjacency matrices, quotient matrices and WL-closures of these graphs.

Thanks for your attention!